Code A

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main(int argc, char *argv[]) {
    int src, dest;
    char buffer[4096];
    ssize_t n;

    if (argc != 3) {
        printf("Usage: %s source destination\n", argv[0]);
        return 1;
    }

    src = open(argv[1], O_RDONLY);
    if (src == -1) {
        perror("Source open failed");
        return 1;
    }

    dest = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (dest == -1) {
        perror("Destination open failed");
        close(src);
        return 1;
    }

    while ((n = read(src, buffer, sizeof(buffer))) > 0) {
        ssize_t total = 0;

        while (total < n) {
            ssize_t w = write(dest, buffer + total, n - total);
            if (w <= 0) {
                perror("Write failed");
                close(src);
                close(dest);
                return 1;
            }
            total += w;
        }
    }
```

```
    if (n < 0)
        perror("Read failed");

    close(src);
    close(dest);

    if (n < 0)
        return 1;

    printf("File copied successfully.\n");
    return 0;
}
```

```
adithya@LAPTOP-07A78KDI:~/OS/OS$ gcc lowcopy.c -o lowcopy
adithya@LAPTOP-07A78KDI:~/OS/OS$ echo "Hello Linux Programming" > input.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ ./lowcopy input.txt output.txt
File copied successfully.
adithya@LAPTOP-07A78KDI:~/OS/OS$ cat output.txt
Hello Linux Programming
adithya@LAPTOP-07A78KDI:~/OS/OS$
```

1B

```
#include <stdio.h>

int main(int argc, char *argv[]) {
    FILE *src, *dest;
    char buffer[4096];
    size_t n;

    if (argc != 3) {
        printf("Usage: %s source destination\n", argv[0]);
        return 1;
    }

    src = fopen(argv[1], "rb");
    if (src == NULL) {
        perror("Source open failed");
        return 1;
    }

    dest = fopen(argv[2], "wb");
```

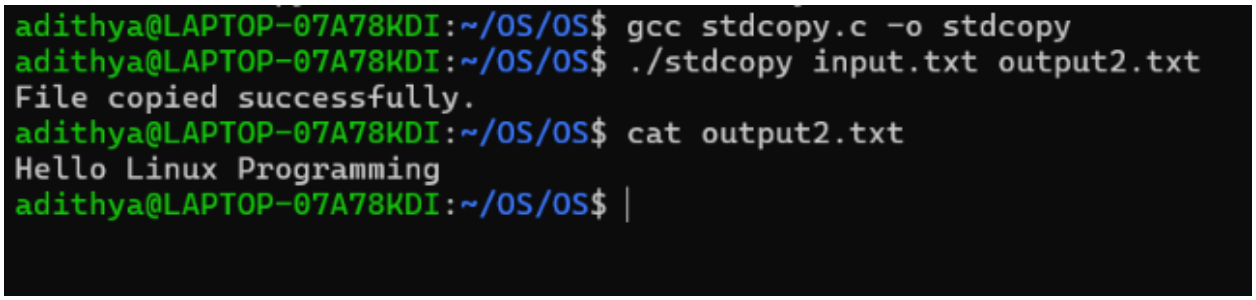
```
    if (dest == NULL) {
        perror("Destination open failed");
        fclose(src);
        return 1;
    }

    while ((n = fread(buffer, 1, sizeof(buffer), src)) > 0) {
        if (fwrite(buffer, 1, n, dest) != n) {
            perror("Write failed");
            fclose(src);
            fclose(dest);
            return 1;
        }
    }

    if (ferror(src)) {
        perror("Read failed");
        fclose(src);
        fclose(dest);
        return 1;
    }

    if (fclose(src) != 0 || fclose(dest) != 0) {
        perror("Close failed");
        return 1;
    }

    printf("File copied successfully.\n");
    return 0;
}
```



Program 2

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main(void) {
    int in, out;
    char buffer[100];
    ssize_t n;

    in = open("input.txt", O_RDONLY);
    if (in == -1) {
        perror("Input open failed");
        return 1;
    }

    out = open("output.txt",
            O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (out == -1) {
        perror("Output open failed");
        close(in);
        return 1;
    }

    if (dup2(in, STDIN_FILENO) == -1) {
        perror("Input redirection failed");
        close(in);
        close(out);
        return 1;
    }

    if (dup2(out, STDOUT_FILENO) == -1) {
        perror("Output redirection failed");
        close(in);
        close(out);
        return 1;
    }

    close(in);
    close(out);

    while ((n = read(STDIN_FILENO, buffer, sizeof(buffer))) > 0) {
```

```
        ssize_t total = 0;

        while (total < n) {
            ssize_t w = write(STDOUT_FILENO,
                              buffer + total, n - total);
            if (w <= 0) {
                perror("Write failed");
                return 1;
            }
            total += w;
        }
    }

    if (n < 0) {
        perror("Read failed");
        return 1;
    }

    return 0;
}
```

```
adithya@LAPTOP-07A78KDI:~/OS/OS$ nano redirect.c
adithya@LAPTOP-07A78KDI:~/OS/OS$ gcc redirect.c -o redirect
adithya@LAPTOP-07A78KDI:~/OS/OS$ echo "Linux file descriptor redirection" > input.txt
adithya@LAPTOP-07A78KDI:~/OS/OS$ ./redirect
adithya@LAPTOP-07A78KDI:~/OS/OS$ cat output.txt
Linux file descriptor redirection
adithya@LAPTOP-07A78KDI:~/OS/OS$
```